

02-Light-Emitting-Diodes

July 6, 2026 10:02 PM

Light
Bulbs

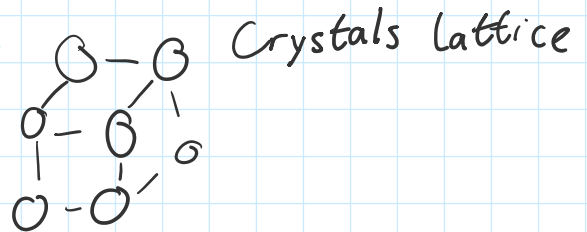
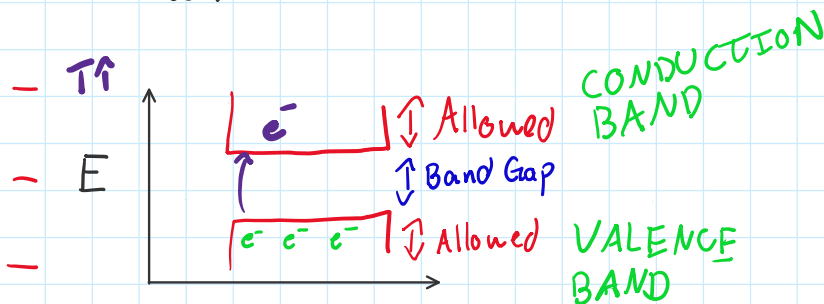


Gets Hot $\rightarrow$ Light

Issue: Generates Heat mostly
95%, inefficient & expensive

Solution: LEDs

Semi-conductors: between insulator & conductor as a material. Can sometimes act as very good insulator or conductor. Depending on signal can change between & be microcontroller; 0-1.
ex. Silicon

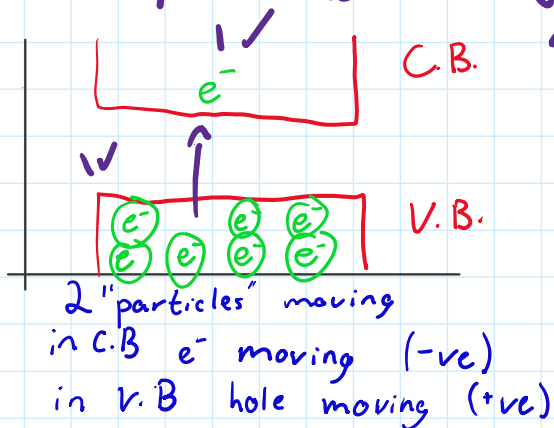


Crystals Lattice

As atoms come close, Pauli exclusion "smears" allowing for regions of e^- instead discrete states.

No current can flow in either band

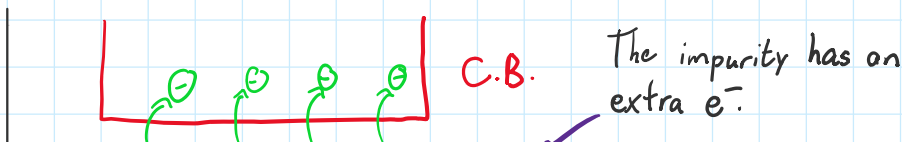
If temp increases electron jumps & allows for flow & current

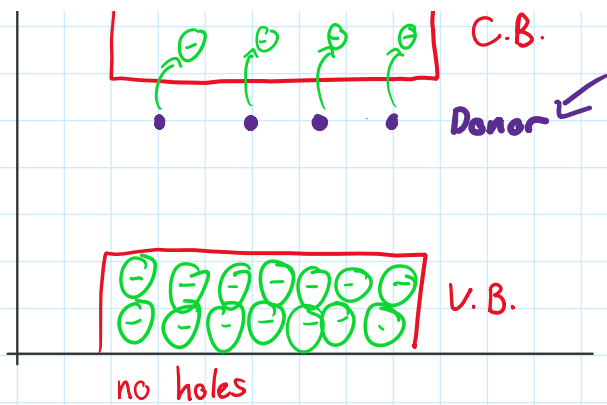


As temp $\uparrow$
Conductivity $\uparrow$

Intrinsic

What if we introduce impurities into the bands?

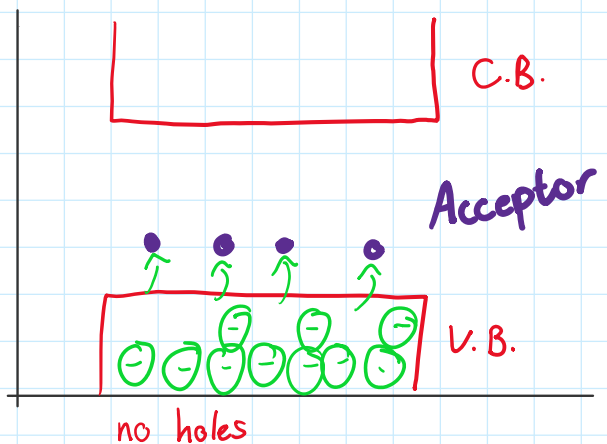




The impurity has an extra e^- .

C.B. is conductor now
V.B. is still insulator

n-type

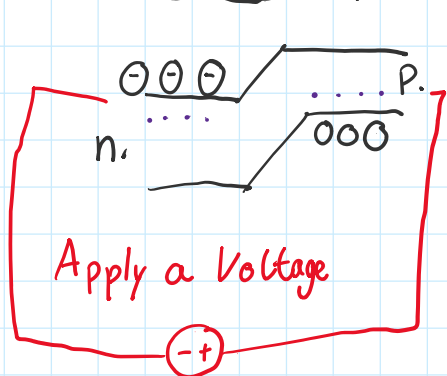


Now impurities are missing e^-

V.B. now conducts (by holes)
C.B. insulator

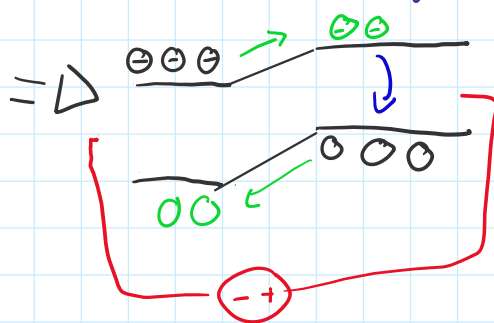
P-type

Connection to LEDs



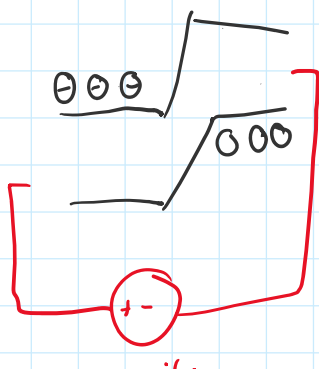
Nothing can happen.

Less steep now



Electron Hole Pair
Recombination

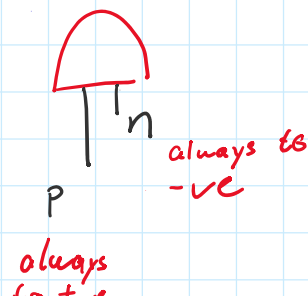
- High E to low E
- ΔE emitted as a Photon (light)
- Higher V $\rightarrow$ more EHP $\rightarrow$ more photons



No current & no light

$\therefore$ +ve to p-type
-ve to n-type

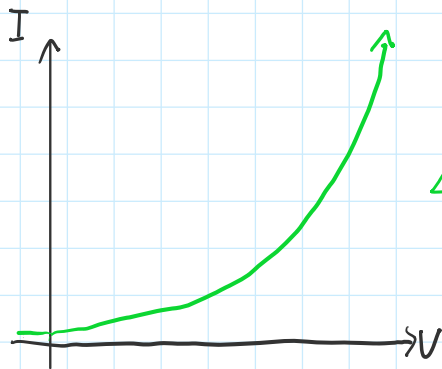
$\therefore$ must be wired correctly



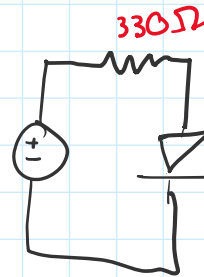


∴ must be wired correctly

always
to +ve

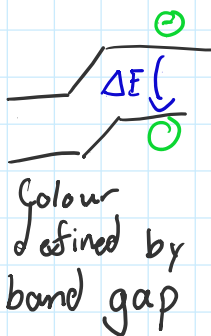


exponential
 ↳ can lead to heat
 & burning out
 ↳ needs a resistor in
 series



How is colour determined?

Determined by ΔE & ∴ λ of emitted Photon



$$\Delta E = E_p = hf$$

$$f = \frac{1}{\lambda}$$

$f \uparrow \rightarrow$ purple
violet
uv

$f \downarrow \rightarrow$ red
infrared